

Chapter 7

Hebrides English in the 1980s and now

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Hebrides English is a distinct Scottish Gaelic-influenced variety of English spoken in the Hebridean island chain. The variety is most likely of relatively recent origin: before the second half of the nineteenth century, English was not widely spoken in this part of Scotland, where until recently Gaelic has been the dominant language. Hebrides English exhibits a variety of distinctive linguistic features which may be due to widespread bilingualism among its speakers in both Gaelic and English, including preaspiration and sibilant devoicing. The present study examines these two features, as well as glottalization, a possible incoming feature, in previously unanalyzed archival material from the early 1980s collected on the islands of Lewis, Harris, and Skye. The chapter thus offers a glimpse of Hebrides English as it existed four decades ago, and taken in context with more recent research (e.g. Clayton 2017, 2018) allows further insight into a little-studied variety of English evidently in considerable flux.

1 Introduction

1.1 Outline of this study

This chapter has two chief goals. The first is to offer some historical perspective on the phonological characteristics of Hebrides English, based on archival data collected in the early 1980s, and in so doing help to develop a picture of Hebrides English as it was spoken forty years ago. The second goal is to point to certain changes that have taken place in Hebrides English in the intervening decades.



The chapter makes use of two main sources of data. The first is the large body of sociolinguistic interviews conducted by Cynthia Shuken in the early 1980s.¹ Much of the data from that collection of material has never previously been analyzed or published (Shuken 1984, 1985, and 1986 offer analyses of subsets of the material). Shuken recorded interviews with nearly 100 individuals from Lewis, Harris, and Skye. Upon her departure from academia in the late 1980s, she placed these recordings and associated questionnaires in an archive at the University of Edinburgh School of Scottish Studies, where they have remained effectively untouched since. This collection represents an invaluable and unique record of Hebrides English as it was used in that era and region, and contains ample material to support a much more wide-ranging analysis than current space permits.

The second body of data referred to in this chapter is that collected by the first author during fieldwork in 2013 (cf. Clayton 2015, 2017, 2018). These studies included data from 24 speakers of Hebrides English, representing nine islands: Skye, Harris, Lewis, North Uist, South Uist, Grimsay, Raasay, Tiree, and Vatersay. While the two collections are not perfectly parallel, differing in numbers of participants, geographic scope, and to some extent in methodology, there is sufficient overlap between them to permit a useful examination of the trajectory of two sociophonetic variables over the time period in question: preaspiration and glottalization. The present chapter also considers sibilant devoicing as it occurs in Shuken's data; the Clayton study did not address such devoicing and so allows no comparative analysis of this feature.

The balance of the chapter proceeds as follows: Section 2 describes a number of distinguishing characteristics of Hebrides English, and provides a review of the existing literature on the topic. Section 3 describes the material in the Shuken archive in detail. Section 4 presents the analyses of preaspiration, glottalization, and devoicing, while Section 5 offers a summary of the results and a discussion.

2 Background and prior research

2.1 What is Hebrides English?

The remarkably diverse linguistic landscape found in Scotland is in large part due to the prolonged conditions of linguistic contact that have obtained in the region

¹While the first author is the writer of this chapter and conducted the bulk of the analysis it incorporates, the chapter depends crucially on Cynthia Shuken's very substantial prior work, and indeed was possible only because she made her collected materials available to the first author. Upon request, Dr. Shuken agreed to be identified as a co-author, despite no longer working in academia. Any errors the chapter contains are solely those of the first author.

for centuries. The island groups of Shetland and Orkney to the northeast of the Scottish mainland, for example, are home to a variety of insular Scots influenced by Norn, the North Germanic language once spoken in those islands (Melchers 2008, McClure 1994, Barnes 1984, Wells 1982, Catford 1957a). Meanwhile, in the Highland and Island regions of northwestern Scotland one encounters English varieties influenced to a greater or lesser degree by the lexicon, syntax, and phonology of Scottish Gaelic, the Celtic language that has been, until the last few decades, the dominant language in the region (McClure 1994, Shuken 1984, 1985, Clement 1997, Stuart-Smith 2008, Clayton 2017, 2018).

While these Gaelic-influenced varieties are sometimes referred to collectively as *Highland and Island English*, they are by no means uniform, and further subdivisions are possible. Frequently, for example, a distinction is made between Highland English on the one hand, and Hebrides (or Hebridean or Island) English on the other (Clement 1980, 1997, Shuken 1985, Sabban 1985, Bird 1997). The basis of this division is the degree and kind of presumed contact with Gaelic. Thus by *Highland English* is usually meant the form spoken in the mainland Highland regions more or less adjacent to Scots-speaking areas. In this part of the Highlands, Gaelic has not been widely spoken for many decades if not longer, so linguistic contact between Gaelic and English is more a matter of historical record than contemporary fact. Hence, such characterizations of Highland English as “standard English on a Gaelic substratum” (Mather & Speitel 1986), or as “post-Gaelic English” (Catford 1957b) are reasonable ones.

By contrast, *Hebrides English* refers self-evidently to the variety of Scottish English frequently heard in the Hebridean island chain, especially those islands where Gaelic has at least until very recently remained a widely used language, and where bilingualism is commonplace if not universal. Though (Catford 1957b’s) description of Hebrides English (HE hereafter) as “the English spoken, with varying degrees of imperfection, as a second language, by persons whose mother tongue is still Gaelic” is difficult to sustain six decades on as usage of Gaelic in the area continues to decline (Ó Giollagáin et al. 2020), it is nevertheless true that the degree of contact between English and Gaelic is much closer in these islands than anywhere else in Scotland.

Another factor likely to be relevant is the manner in which influences from English, or from Scots, have made their way into formerly Gaelic-dominant communities. Clement’s (1980) point that learning English at school is quite distinct from learning Scots from one’s neighbors remains a perspicuous one. English was largely introduced to the islands and more remote parts of the Highlands via instruction in the schools, upon English being made compulsory by the Education Act of 1872 (Withers 1984). By the 1970s, bilingualism in English was

effectively universal among Gaelic speakers, though even today one can readily find native Gaelic speakers who knew little or no English before beginning primary school. Thus, it is probably appropriate to regard Hebrides English as a variety of schoolroom Scottish Standard English that features significant Gaelic influences.

This description of HE, however, does not account for recent influences from urban varieties of Scots or Scottish English. The larger Island communities, such as Portree on Skye, and Stornoway on Lewis are prominent commercial centers with significant numbers of English- and Scots-speaking incomers with concomitant influences on local language use (Shuken 1984, Speitel 1981, Clement 1980). These influences seem to include the introduction of velarized [ɫ] (Shuken 1985) and glottal reinforcement (Clayton 2018, Wells 1982, Trudgill 1988, Stuart-Smith 1999a,b); there may well be others.

Finally, it must be stressed that English as used in the Hebrides is hardly uniform, as this chapter and other papers illustrate. What research has been published, largely focuses on the northern islands of the Hebrides, especially Skye, Harris, North Uist, and Lewis, and therefore presents a very incomplete picture. There remains a marked absence of research regarding English as it is spoken on the other Hebridean islands.

2.2 Possible Gaelic influences in Hebrides English

Discussions of HE in the literature frequently mention certain features that may well be due to contact with Gaelic. They include:

2.2.1 Morphosyntax

1. The “BE after V-ing” construction (Sabban 1985):

- (1) Tha mi dìreach air mo nigheadaireachd a’ dhèanamh
be.PRS I just PERF my washing PROG doing
‘I’m just after finishing my washing.’

2. Absence of the English present perfect construction (Sabban 1985):

- (2) Tha mi air a bhith nam bhantrach-fhìr a-nis airson sia
be.PRS I PERF ART be.VN in-my widower now for six
bliadhna
years
‘I’m a widower now for six years.’

3. Cleft constructions (Shuken 1984):

- (3) ‘S e Gàidhlig a bhruidhinn sinn an còmhnaidh aig an
 cop.PRS it Gaelic REL spoke we always at the
 taigh
 home
 ‘It was always Gaelic we spoke in the home.’

2.2.2 Phonology

1. There is no voicing contrast among Gaelic sibilants. HE realizations like *measure* as [mɛʃur], and *just* as [tʃʌst] are thus presumably due to Gaelic influence (Shuken 1985, Wells 1982).
2. In the Hebridean varieties of Gaelic, the unaspirated voiceless stops /p t k/ contrast with the voiceless aspirated series /p^h t^h k^h/. There are no voiced stops. In like vein, many HE speakers omit voicing in English /b d g/, e.g. *glory* [klɔri], *ragged* [rækit], *building* [pɪltɪŋ] (Grant 1913, Borgstrøm 1940, Oftedal 1956, Wells 1982, Ladefoged et al. 1998, Gillies 2009).
3. In many varieties of Gaelic, the aspirated stop phonemes /p^h t^h k^h/ are preaspirated in non-initial positions, for instance as [p^h t^h k^h] (the typical Lewis pattern), or as [p^h t^h xk] (as in Skye and Harris). In similar fashion, voiceless stops in HE may feature preaspiration: *top* [t^hɔp], *not* [nɔt], *cat* [k^hæt], *bake* [be^hk] (Borgstrøm 1940, Oftedal 1956, Shuken 1979, Ní Chasaide 1985, Ladefoged et al. 1998, Clayton 2010, Nance & Stuart-Smith 2013). Lewis speakers have been particularly singled out in this regard, e.g. Borgstrøm (1940) notes that “postvocalic *k*, *t*, *p* are usually rendered by Lewismen as ^h*k*, ^h*t*, ^h*p*, and nobody seems to be aware that this is not the proper English pronunciation” (p. 21). Likewise, Shuken (1984, 1985) found abundant preaspiration in Lewis English, but rather less among Skye and Harris speakers.
4. Epenthetic vowels: In Gaelic, epenthesis affects post-vocalic underlying VC₁C₂ sequences: the vowel must be short, C₁ must be a nasal or liquid, and C₂ must be a non-homorganic consonant apart from /p^h t^h k^h/. In such instances, an epenthetic vowel will be inserted between the two consonants and which matches the vowel preceding the CC cluster, e.g. *arm* /arm/ ‘army’ becomes [a.ram], *airgead* /ɛr^hk^hæt/ ‘money’ becomes [ɛr^hɛk^hæt]. In HE, a similar, though not identical, pattern obtains among some speakers, though it seems to apply in slightly different phonological conditions:

an epenthetic vowel may appear in VC₁C₂ sequences in which C₁ is a nasal or liquid, just as in Gaelic, though in HE the CC may be homorganic: *milk* [mɪlɪk], *farm* [faram], *kiln* [kɪlɪn] (Hammond et al. 2014, Smith 1999, Bosch & de Jong 1997).

5. Dental stops: the coronals /t d/ may be dentalized as in Gaelic, e.g. *tree* [tʰri] (Grant 1913, Holmer 1957, Dorian 1978, Holmer 1981, Shuken 1984, Ladefoged et al. 1998).
6. Retroflex /s/: HE frequently features retroflexion of /s/ after /r/, as in *first* [fɪɹʂt], *horse* [hɔɹʂ], *parcel* [paɹʂəl], a pattern also found in Gaelic, e.g. *spors* [spo:ɹʂ] ‘sport’, *dearrsach* [tʰaɹʂax] ‘bright’ (Borgstrøm 1940, Wells 1982, Shuken 1984, Stuart-Smith 2008).
7. Laterals: The lateral liquid common among HE speakers is a clear [l]. Elsewhere in Scotland, a velarized [ɫ] frequently predominates. While some varieties of Gaelic do have both velarized and non-velarized laterals, the velarized variant is rare in the northern Hebrides (Holmer 1957, 1981, Ternes 1973, Dorian 1978, Wells 1982, Shuken 1984, 1985, Ó Murchú 1989, Johnston 1997, Ladefoged et al. 1998, Stuart-Smith 1999a, Nance 2014).

2.3 Prior research

2.3.1 Published studies of Hebrides English

Published in-depth studies of Hebrides English are notable for their scarcity. Those that are available point both to the distinctive character of the variety (in particular those features potentially of Gaelic origin, as described above), and to progressive convergence with mainland varieties through a process of dialect leveling, or what Hickey (2013) refers to as supraregionalisation: a process in which a particular language variety loses those identifiably local features that have previously set it apart from a more broadly distributed (and perhaps less stigmatized) variety. Below I describe the work of Sabban (1985), Shuken (1984, 1985, 1986), and Clayton (2015, 2017, 2018). I delve into the Clayton and Shuken studies in fair detail because they are the most relevant to the present chapter, particularly Shuken’s work.

2.3.2 Sabban (1985)

Sabban (1985) is a morphosyntactic study of the English spoken in the Hebridean islands of Skye and North Uist. It is to date the only published study of HE morphosyntax, so far as I am aware. Sabban examined two patterns: the *BE after*

V-ing construction, and the absence of the present perfect (cf. Section 2.2). Participants included 29 bilingual speakers, aged 6 to 89. Sabban found that these two patterns were overwhelmingly concentrated in speakers over 50, while speakers under 20 used them very rarely if at all. These results strongly suggest that these morphosyntactic features were obsolescent in these areas at the time of her study, a situation that is likely only to have advanced in the intervening four decades.

2.3.3 Shuken (1984, 1985, 1986)

In the late 70s and early 80s, Cynthia Shuken conducted the first systematic sociophonetic study of Hebrides English (1984, 1985, 1986), with the aims of providing a thorough phonetic description of the variety, and identifying any key sociolinguistic variables that might be functioning as indices of identification with socioeconomic groups. She recorded nearly one hundred Hebridean English speakers, representing the islands of Harris, Lewis, and Skye. Her material therefore is confined to the English spoken in these three islands. The participant pool was balanced between male and female speakers, and grouped into three age brackets (17–25, 35–45, and 55+). In addition, participants included rural speakers from all three islands, and urban speakers drawn from Stornoway, the chief town on Lewis; and from Portree, the principal population center on Skye. The Skye and Lewis urban speakers were further subdivided into English monolinguals and English-Gaelic bilinguals. Since no comparable urban center exists on Harris (its largest community, Tarbert, had a population of 464 in the 1981 census), Shuken's 14 Harris speakers were all categorized as rural bilinguals. At the time of the study, the rural areas of all three islands were overwhelmingly bilingual, and so she made no effort to include rural monolinguals. Her collected material included recordings of wordlists, reading passages, and unstructured conversation.

Three publications resulted from this study. The first, Shuken (1984), provides a brief history of the variety as well as a profile of the phonology and morphosyntax of HE. No quantitative analysis is presented in this paper.

The second publication, Shuken (1985), was a sociophonetic analysis of HE based on data from the youngest age bracket in her participant group, i.e. 17–25 year olds, including rural bilinguals, urban bilinguals, and urban monolinguals. This paper compared the relative abundance of two HE variables of potential Gaelic provenance, the unvelarized or clear [l] and the preaspiration of voiceless stops, with two variables of likely external origin, namely the velarized lateral [ɫ]

and the glottalization of voiceless stops (both in the form of glottal replacement or T-glottaling, and glottal reinforcement).

Shuken found that the use of velarized [ɫ] rather than clear [l] was much more abundant among urban speakers compared to rural speakers in both Skye and Lewis, much more abundant on Skye overall than Harris or Lewis, and most abundant among urban monolinguals from Skye. Conversely, clear [l] dominated among rural speakers, especially those from Lewis.

Shuken found that preaspiration was more common among Lewis speakers than those from Harris or Skye, that it was slightly more common among rural speakers than urban speakers in Lewis, and that it was of similar frequency in both rural and urban speakers in Skye. On the other hand, she found that glottalization was much more abundant among Skye speakers than those from Harris or Lewis, and more abundant among urban speakers than rural speakers in both Lewis and Skye. Finally, she found that glottal replacement was more common among Skye speakers than those from Harris or Lewis and more common among urban speakers than rural speakers.

Given the greater relative abundance of velarized [ɫ] and glottalized forms among young speakers in urban communities, and among those in greater proximity to the mainland, Shuken concluded that there was evidence for a significant and growing degree of mainland influence on HE at the time of this study.

Shuken (1986) examined the phenomenon of vowel lengthening among HE speakers in relation to the patterns seen in Gaelic, in Scottish Standard English, and in Received Pronunciation. She found that a small number of word pairs like *great* ~ *grate* and *maid* ~ *made* were in contrast among some speakers, such that one member of the pair reliably differed from the other in vowel length. Shuken suggested that the feature is potentially due to influence from Gaelic, where contrastive vowel length is pervasive, e.g. *srac* [s̪rɐxk] ‘rend!’ versus *stràc* [s̪t̪rɐ:xk] ‘stroke’. However, in HE such unconditioned contrasts were seen to be sporadic, and with no consistent pattern between speakers.

Apart from the few instances of contrastive lengthening, Shuken found two patterns of conditioned vowel length which varied in frequency by island. Speakers from Harris and Lewis were most likely to follow the pattern widely observed in Scottish Standard English, wherein (most) vowels are predictably lengthened before voiced fricatives, /r/, and morpheme boundaries (Aitken 1981). Meanwhile, Skye speakers of HE were more likely to follow the Received Pronunciation pattern of lengthening vowels predictably before voiced consonants and in open syllables. In other words, the RP pattern had a greater degree of influence in Skye, the island closest to the mainland, than in either of the two more remote islands.

In general, then, Shuken's published work on HE points to a linguistic situation in considerable flux, with evidence of expanding influence from mainland varieties of English.

2.3.4 Clayton (2015, 2017, 2018)

The only work on HE to be published since the mid-1980s is found in Clayton (2015, 2017, 2018). This study surveyed 24 bilingual speakers of HE and Gaelic, 10 male and 14 female. A range of islands within the Hebrides archipelago were represented, though unfortunately it was extremely difficult to identify more than one or two speakers from some of the islands. These islands included Skye (8 speakers), Harris (1), Lewis (8), North Uist (2), South Uist (1), Grimsay (1), Raasay (1), Tiree (1), and Vatersay (1). The study addressed the incidence of preaspiration and of glottalization (both as glottal reinforcement and as T-glottaling). Preaspiration was found to be abundant only among a subset of older female speakers from Lewis, suggesting both that the feature is concentrated in the English of Lewis (cf. Borgström 1940, Shuken 1984, Sabban 1985), and that preaspiration is falling out of use among younger speakers. Glottal reinforcement was most abundant in younger speakers, particularly younger female speakers, though no clear pattern was found indicating that these features were comparatively more abundant in Skye than in Harris or Lewis (cf. Shuken 1985 with a similar result). Finally, T-glottaling was highly restricted in its occurrence: it only occurred in the word *rotten*, and predominantly among Skye speakers. Neither age nor sex appeared to affect the frequency of T-glottaling. The greater abundance among younger speakers suggests that glottal reinforcement, well-established in urban varieties of Scottish English (Stuart-Smith 2008, 2003, 1999a,b, Wells 1982), are advancing in Hebrides English as well.

3 The Shuken archive

3.1 Overview

Cynthia Shuken's collected materials are archived at University of Edinburgh's School of Scottish Studies. They include about 85 tape reels, of which 40 contain sociolinguistic interviews relevant to HE; detailed IPA transcriptions of some of the recorded interviews; 92 completed sociolinguistic questionnaires; and an assortment of fieldwork notes.

The tapes in the archive seem to be copies in most instances rather than originals. Often, material from one speaker is scattered across several tape-reels. In a

small number of cases, identical material occurs on more than one tape. While the tape-reels are in labeled boxes, in most cases the labels correspond only partially, or else not at all, with the material actually contained on the tape. These factors presented significant organizational challenges which had to be overcome before the present analysis could commence.

All tape-reels in the collection were digitized by the very helpful archivists at the first author's request; these digitized versions were used for all the new analysis presented in this chapter.

3.2 Recording procedures

While a small number of Shuken's participants were recorded in a laboratory setting at the University of Edinburgh, most were recorded in the field using a portable Uher reel-to-reel tape recorder and a table-top Sennheiser microphone. Recording locations varied widely, and included the participants' homes, workplaces, and local primary schools among other settings. Most recordings were made in quiet conditions, but given the nature of fieldwork, some recordings inevitably feature a degree of ambient noise, such as doors slamming, background conversations, vehicle engines idling, and so forth. In no case, however, did such background noises significantly impede the type of analysis carried out for this chapter. The exact model and specifications of the recorder and microphone are unavailable at this point, but it seems safe to assume that they were industry standard for the era. The overall audio quality of the recorded material is generally quite high.

During a typical recording session, Shuken would first ask the participant to state their name and place of residence. She would then have them read two passages in English. Gaelic-speaking participants would then read a list of Gaelic words, and then move on to the English word list. All words were given to participants as printed lists without using a frame sentence; all participants encountered identical lists. Usually, Shuken would conclude each session by having participants complete a sociolinguistic questionnaire; frequently, the recording includes the participant's commentary on the questionnaire, though not always.

3.3 Questionnaires

Shuken used a detailed 7-page sociolinguistic questionnaire to collect demographic information about her participants as well as their cultural and political attitudes. Two versions of the questionnaire were prepared: one for the use of Gaelic-speaking bilinguals, and one for English-speaking monolinguals. Among

other things, the questionnaire asked about each participant's place of birth and that of their parents; their residence history, education and employment; their degree of personal identification with the local community and region, with Scotland, and with the United Kingdom; their preferences and degree of knowledge of cultural matters such as traditional Highland music, literature, and history; the frequency with which they spoke Gaelic (if bilingual) and with whom; their perceptions of the relative attractiveness and correctness of various dialects of Gaelic and English; and their degree of pride in speaking a distinct variety of English. The first page of the bilingual version of the questionnaire is reproduced in Figure 1 (page 258).

The archive contains completed sociolinguistic questionnaires for 92 individuals: 46 from Lewis, 14 from Harris, and 32 from Skye. While the questionnaires are numbered consecutively (L1-L50 for Lewis, H1-H14 for Harris, and S1-S50 for Skye), there are gaps for 4 questionnaires from the Lewis group and 18 from the Skye group, suggesting that the corresponding recordings were not made. Recorded interviews corresponding to almost all of the completed questionnaires could be located on the tape reels, with the exception of three participants from Lewis, one from Harris, and three from Skye; however, Shuken's notes include complete, highly detailed transcriptions of the English wordlist and the two reading passages for one of the three missing Skye speakers (S12), and for the Harris speaker (H6). Thus, there is audio available for 85 interviews plus detailed transcripts of two more interviews.

3.4 Participants

Shuken's full participant pool includes 92 speakers of HE from the islands of Lewis, Harris, and Skye, as represented in the archive (if we include the seven participants for whom we have questionnaires but no recordings). The pool was designed to be evenly balanced for sex and was stratified by age group (17–25, 35–45, 55+). In addition, Shuken included rural speakers from all three islands as well as urban speakers drawn from Stornoway, the chief town on Lewis, and from Portree, the principal population center on Skye. The urban speakers included both monolingual HE speakers and bilingual speakers of HE and Gaelic.

A bit more could be said at this point about the towns of Stornoway (on Lewis) and Portree (on Skye). At the time of Shuken's study, the parish of Stornoway had a population of approximately 7900, according to the 1981 census (Shuken 1985). The town's economy was, and still is, centered on retail, social services, tourism, textiles, aquaculture, and offshore gas and oil extraction (Highlands &

GEBQ-1 SSSA/58/13/9

LOCALITY: _____

Name: _____

Date of Birth: _____

Place of Birth: _____

Sex: Male _____ Female _____

Occupation: _____

Length of Residence in Locality: _____

Where else have you lived? For how long? _____

Primary School: _____

Secondary School: _____

Further education, if any: _____

Mother from: _____

Father from: _____

If married, spouse from: _____

If married, does spouse speak Gaelic? Yes _____ No _____

1. How would you describe yourself? Pick the three most important, and label them first, second, and third most important (1, 2, 3):

- a.) as a Highlander _____
- b.) as a Briton _____
- c.) as a Gael _____
- d.) as a member of your community _____
- e.) as an Islander _____
- f.) as a Scotsman/Scotswoman _____
- g.) as a Lewis/Harris/Skye person _____

2. Do you enjoy the following? How much?

	A lot	A little	Not at all
a.) music from the Mòd _____			
b.) country-western music _____			
c.) traditional Gaelic songs _____			
d.) plobaireachd _____			
e.) céilidhs _____			
f.) discos _____			
g.) Scottish country dancing _____			
h.) classical music _____			
i.) céilidh bands _____			
j.) Irish music _____			
k.) Welsh music _____			
l.) clàrsach music _____			
m.) Highland games _____			
n.) dance music played on pipes _____			
o.) pennywhistle _____			

Figure 1: Page from Shuken's Gaelic-English bilingual questionnaire

Enterprise 2020). Stornoway's population has declined somewhat since 1981; the estimated population as of 2020 was 7280 (National Records of Scotland 2023).

Like Stornoway, Portree is a regional commercial center, though more heavily dependent on tourism and arguably more subject to outside influence as a consequence. The completion of a road bridge connecting Skye to the mainland in 1995 has only strengthened the commercial and social ties between Skye and the mainland since Shuken conducted her fieldwork (Highlands & Enterprise 2007) and thus magnified the potential for mainland linguistic influence. The 1981 census recorded a population of 1414 in Portree compared to an estimated 2,310 in 2020 (National Records of Scotland 2023), indicating a significant pattern of growth in the community.

Shuken was able to record participants representing nearly every possible combination of demographic factors in her planned speaker population. For most combinations of the factors of island, age, rural/urban, and monolingual/bilingual, her recorded interviews include at least one male and one female speaker. The majority achieved her target of at least two male and two female speakers for each of these conditions. Where unfilled gaps do occur, these include female urban monolinguals from Lewis in the 55–65 age group, male and female rural bilinguals from Skye in the 17–25 age group, Skye rural bilingual males in the 35–45 age group, and Skye urban monolingual males in the 55–65 age group; no speakers representing these subgroups are present in the recorded material. Conversely, in a small number of conditions “surplus” speakers were recorded, i.e. more than the target of two women and two men per condition. For the present study, the data from only two male and two female speakers per condition were included in the analysis, in order to prevent overrepresenting any subpopulation.

The final speaker pool for the present study contains 70 individuals, with the distribution shown in Table 1.

3.5 Stimuli and reading passages

Shuken's full stimulus set consists of 319 English words.² While the majority of the stimuli presented no difficulties for participants, the recordings do indicate that several of the words in the list were unfamiliar to many people, who struggled to read them accurately as a consequence. Problematic words included *quoit*, *adroit*, *gesture*, *loathe*, and *rouge*. These words were excluded from the present analysis. In addition, the word *use* was excluded because it was orthographically ambiguous: participants were unsure whether the verb (with a final /z/)

²The full set of stimuli is available upon request to the first author.

Table 1: Distribution of participants according to island (Lewis, Skye, Harris), age group, sex, and urban/rural, bilingual/monolingual (e.g. LRB = Lewis, rural, bilingual)

Age/sex	LRB	LUB	LUM	SRB	SUB	SUM	HRB
17–25 F	L1 L2	L13 L14	L25 L26	-	S1	S25 S26	H1 H2
17–25 M	L3 L4	L15	L27 L28	-	S3 S15	S27 S28	H3 H4
35–45 F	L5 L6	L17	L29 L30	S37 S38	S5 S6	S48	H5 H6
35–45 M	L7 L8	L19 L20	L31 L32	-	S7 S19	S31 S49	H7 H8
55+ F	L9 L10	L21 L22	-	S9 S10	S40 S42	S33 S34	H9 H10
55+ M	L11 L12	L23 L24	L35 L36	S11 S12	S41 S50	-	H11 H12

was meant, or the noun (with a final /s/), and so the word could not be reliably analyzed.

In addition to the list of individual words, Shuken also presented participants with two short reading passages. Analysis of the data offered by these reading passages is ongoing, and is not included in this chapter. For the three variables considered in the present analysis, the relevant subsets of stimuli are illustrated in Tables 2 and 3.

Table 2: Stimuli for preaspiration and glottalization of /p t k/

/p/	/t/				/k/	
<i>chip</i>	<i>about</i>	<i>boot</i>	<i>grate</i>	<i>soot</i>	<i>actually</i>	<i>plucking</i>
<i>cup</i>	<i>bait</i>	<i>booty</i>	<i>great</i>	<i>suit</i>	<i>choke</i>	<i>spoke (n.)</i>
<i>dip</i>	<i>beat</i>	<i>caught</i>	<i>idiot</i>	<i>water</i>	<i>creak</i>	<i>spoke (v.)</i>
<i>loop</i>	<i>beauty</i>	<i>coat</i>	<i>illuminate</i>		<i>creek</i>	<i>tack</i>
<i>ship</i>	<i>beet</i>	<i>cot</i>	<i>mate</i>		<i>hawk</i>	<i>taxes</i>
<i>staple</i>	<i>bet</i>	<i>crate</i>	<i>petal</i>		<i>joke</i>	<i>taxis</i>
<i>stoop</i>	<i>bit</i>	<i>cut</i>	<i>put</i>		<i>leak</i>	<i>weak</i>
<i>stripes</i>	<i>bite</i>	<i>feature</i>	<i>quite</i>		<i>leek</i>	<i>week</i>
<i>tip</i>	<i>boat</i>	<i>foot</i>	<i>right</i>		<i>lock</i>	

Table 3: Stimuli used to evaluate voicing contrasts in sibilants

	final /z/	initial /z/	final /dʒ/	medial /dʒ/	medial /ʒ/
<i>boys</i>	<i>rays</i>	<i>zeal</i>	<i>badge</i>	<i>dredges</i>	<i>measure</i>
<i>cause</i>	<i>resume</i>		<i>cadge</i>		
<i>caws</i>	<i>rose</i>				
<i>days</i>	<i>rouse</i>				
<i>daze</i>	<i>rows (boat)</i>				
<i>enthuse</i>	<i>rows (fights)</i>				
<i>ewes</i>	<i>seas</i>				
<i>fez</i>	<i>seize</i>				
<i>fuzz</i>	<i>sighs</i>				
<i>halves</i>	<i>size</i>				
<i>noise</i>	<i>stretches</i>				
<i>ooze</i>	<i>ways</i>				
<i>presume</i>	<i>weighs</i>				
<i>raise</i>					

4 Analysis

4.1 Analysis of preaspiration

In Gaelic, the realization of preaspiration tends to vary depending both on the dialect in question and the affected consonant. In many dialects, particularly those of the eastern mainland now moribund or extinct, preaspiration does not occur at all. Among the varieties of Gaelic found in Lewis, Harris, and Skye, there are two main preaspiration patterns. On the one hand is the Lewis pattern, in which preaspiration is typically realized as a period of glottal frication or breathy voice [VhC ~ VfiC] occurring between a vowel and a following voiceless stop. On the other hand, we have the pattern found in Harris and Skye, where before labial and coronal stops preaspiration appears much as in Lewis, i.e. as glottal frication or breathy voice, but before dorsals it is typically realized as homorganic oral frication, i.e. [çç xk] (Nance & Stuart-Smith 2013, Clayton 2010, Ladefoged et al. 1998, Ó Dochartaigh 1994-1997, Ní Chasaide 1985, Borgstrøm 1940). The oral form [xp xt xk] alone is typical of certain dialects, particularly in Perthshire and Argyll (Iosad et al. 2015, Ó Murchú 1989).

In HE, by contrast, preaspiration seems to be rather less variable. There it typically takes the form of glottal frication or breathy voice (Figure 2). Homorganic

oral frication does infrequently occur, particularly with dorsals (Clayton 2018, 2017, 2015, Shuken 1985).

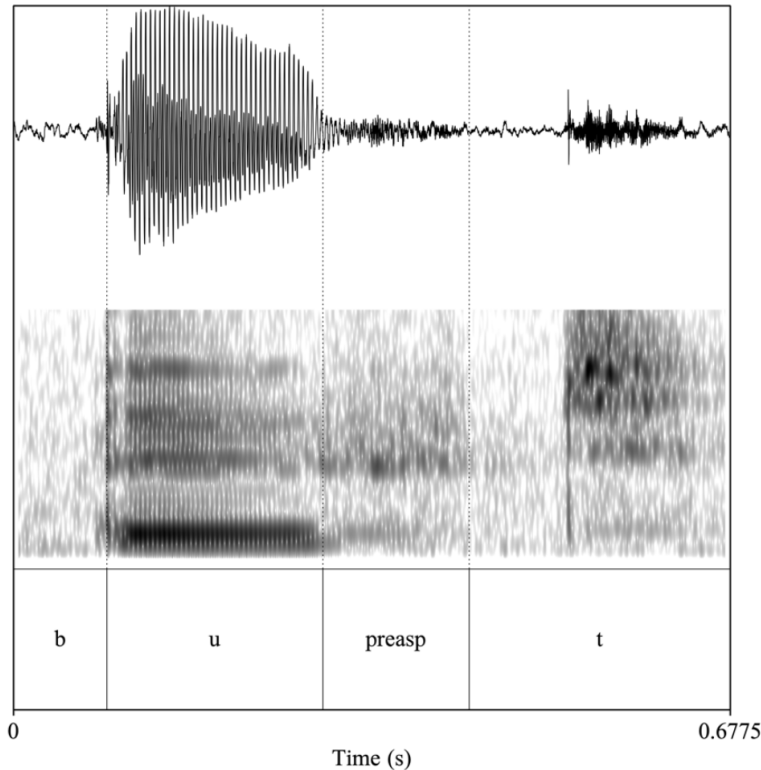


Figure 2: Preaspiration in the word *boot* as produced by L02, a female rural bilingual from Lewis in the 17–25 age group.

For the present study, tokens of preaspiration were identified through inspection of the waveform and spectrogram in Praat (Boersma & Weenink 2020). Of the 319 stimuli produced per speaker, 57 contained the voiceless stops /p t k/ post-vocally in medial or final position, and thus were potential contexts for preaspiration. Of these, 9 stimuli contained /p/, 31 contained /t/, and 17 contained /k/ (Table 2). Totaled across the 70 participants, there were thus 3990 relevant tokens. Of these, 658 tokens (16.5%) were identified as featuring some form of preaspiration (Table 4). Coronal tokens were by far the most numerous, numbering 452 (68.7%) of the total, followed by dorsals (152 tokens, 23.1%), then labials (54 tokens, 8.2%). No labial tokens at all were noted among the Harris and Skye speakers, and very few dorsals (4 out of 29, 13.8%) were present in those two groups. This distribution favoring coronals is similar to that found for HE in Clayton (2017, 2018),

but stands in sharp contrast to the pattern typically observed in Gaelic, where dorsals are favored in terms of frequency (Clayton 2010).

Table 4: Distribution of preaspirated tokens according to island, sex, age group, and place of articulation.

	Lewis		Harris		Skye		
	M	F	M	F	M	F	Total
Age 1 (17–25)							
p	7	13	0	0	0	0	20
t	50	79	1	5	0	0	135
k	15	27	0	0	0	0	42
Total	72	119	1	5	0	0	197
%	25.26	34.80	0.88	4.39	0.00	0.00	
Age 2 (35–45)							
p	6	10	0	0	0	0	16
t	57	81	1	4	2	1	146
k	23	26	0	0	1	0	50
Total	86	117	1	4	3	1	212
%	25.15	41.05	0.88	3.51	1.32	0.44	
Age 3 (55+)							
p	7	11	0	0	0	0	18
t	92	68	3	5	0	3	171
k	28	29	0	0	0	3	60
Total	127	108	3	5	0	6	249
%	37.13	47.37	2.63	4.39	0.00	1.50	
						Total:	658

We see a very uneven distribution of preaspiration according to social factors (Figures 3 and 4). Preaspiration was overwhelmingly localized to Lewis speakers, who produced 629 tokens of the total 658 observed (95.6%). Each of the Lewis speakers produced a minimum of 2 preaspirated tokens, with the exception of L20, an urban bilingual male in the 35–45 age bracket, who produced none.

Statistical analysis of the effects of demographic variables on preaspiration rate (Table 5) was conducted using a logistic mixed-effects model via the `glmer` function of the `lme4` package (Bates et al. 2015, Kuznetsova et al. 2017). Because of the very low rates of preaspiration in Skye and Harris speakers, these islands were excluded from statistical analysis. The final model included *word* as a random effect, and *sex*, *age*, and *rural/urban* as fixed effects. Age was included as

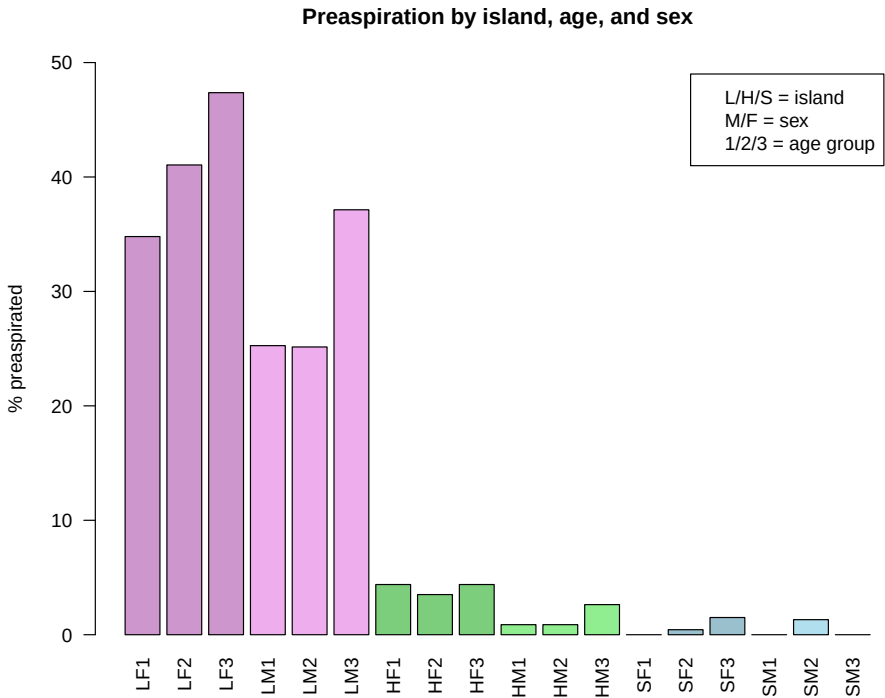


Figure 3: Rates of preaspiration according to island (Lewis, Harris, Skye), sex, and age group (1 = 17–25, 2 = 35–45, 3 = 55+)

a 3-way variable instead of continuous variable, to correspond with the three age bins among Shuken’s participants; in addition, it was not always possible to identify the precise age of some speakers at the time of recording beyond the age bracket they occupied. While *speaker* is commonly included as a random effect in mixed-effect models (Baayen et al. 2008), its inclusion in the models used here tended to obscure the very variability the study hopes to highlight, and so *speaker* was not included as a factor in this and subsequent statistical models.

Among Lewis speakers, the frequency of preaspiration was skewed toward women overall (estimate = -1.389 , $z = -4.039$, $p < 0.001$) (Figure 4c), and toward older speakers (Figure 4b). Older male speakers in particular preaspirated at significantly greater rates than the younger two groups of male speakers (estimate = 1.568 , $z = 3.237$, $p < 0.001$). Urban speakers also preaspirated significantly less than rural speakers (estimate = -1.869 , $z = -6.122$, $p < 0.001$).

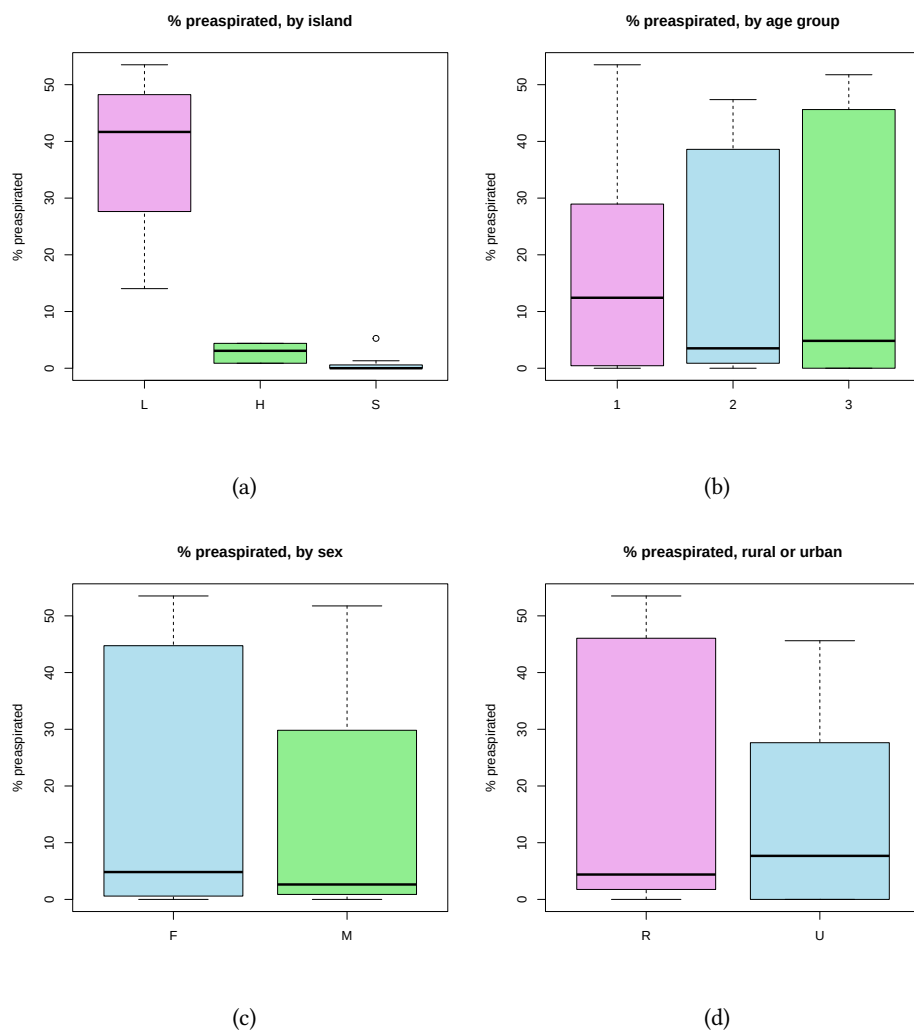


Figure 4: Comparisons of preaspiration rates by island (Lewis, Harris, Skye), age group (1 = 17–25, 2 = 35–45, 3 = 55+), sex, and rural vs. urban.

Table 5: Mixed-effects model of preaspiration rates among Lewis speakers. Intercept represents younger female rural speakers.

Random effects		Variance	SD	
word		4.349	2.085	
Fixed effects	Estimate	SE	<i>z</i>	<i>p</i>
(Intercept)	−0.030	0.373	−0.081	0.936
sexM	−1.389	0.344	−4.039	<0.001***
rurbU	−1.869	0.305	−6.122	<0.001***
age2	−0.591	0.340	−1.738	0.082
age3	−0.301	0.342	−0.880	0.379
sexM:rurbU	1.014	0.441	2.297	0.022*
sexM:age2	1.562	0.481	3.249	<0.001***
sexM:age3	1.568	0.484	3.237	<0.001***
rurbU:age2	1.471	0.430	3.422	<0.001***
rurbU:age3	1.636	0.454	3.607	<0.001***
sexM:rurbU:age2	−3.013	0.627	−4.808	<0.001***
sexM:rurbU:age3	−2.224	0.632	−3.519	<0.001***

4.2 Analysis of glottalization

Just as for preaspiration, glottalization is frequently associated with voiceless stops in HE. Effectively then the number of possible sites for glottalization in the current dataset is 3990, the same as for preaspiration. Of these 3990 tokens, 336 featured some degree of glottalization of the vowel just prior to the stop closure. Outright replacement of the consonant with a glottal stop (T-glottaling) was extremely rare in the Shuken data analyzed for this chapter and so not analyzed separately. Glottalization was identified in the data via irregularities in the timing of vocal fold pulsations (so-called “jitter”), or in the amplitude of the associated waveforms (“shimmer”). Fluctuations in the pitch track provided by Praat could also be observed in some glottalized tokens. Figure 5 illustrates glottalization in the word *leak* as produced by S25, an urban monolingual female speaker from Skye in the 17–25 age group.

Statistical analysis of the glottalization data (Table 7) was conducted by fitting a logistic mixed-effects model via the *glmer* function of the *lme4* package. The final model included *word* as random effect; and *age group*, *sex*, and *bilin-*

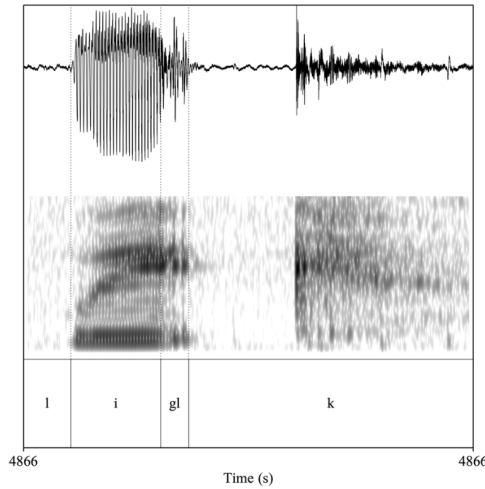


Figure 5: Glottalization in the word *leak* as produced by S25, a female monolingual from Skye in the 17–25 age group.

gual/monolingual as fixed effects. Interactions between the fixed effects were also modeled.

In terms of place of articulation, the distribution of glottalization is considerably more even than preaspiration: 191 of 2170 coronal tokens displayed glottalization (8.8%), 90 of 1190 dorsal tokens (7.6%), and 55 of 630 labial tokens (8.7%). No significant difference by place of articulation was found.

In terms of demographics, however, the distribution of glottalization was in certain respects the inverse of preaspiration (Figure 6). Skye speakers glottalized at far greater rates than Harris or Lewis speakers (Table 6; Figure 7a), glottalizing 251 observed tokens (74.7%), compared to 29 tokens (8.6%) among the Harris speakers and 56 (16.7%) among speakers from Lewis. Age was a significant factor (Figure 7b), where the older two brackets seem to have glottalized at proportionally higher rates, but the differences are very small (age 2: estimate = 1.270, $z = 3.357$, $p < 0.001$; age 3: estimate = 0.922, $z = 2.445$, $p = 0.014$). While women glottalized at slightly higher rates than men (Figure 7c), sex was not a statistically significant factor. Skye women did lead Skye men in the younger two age brackets, though glottalization was more common among older Skye men than women. Outside of Skye, glottalization could be described as abundant only among younger Lewis women; notably, 31 of the 35 tokens attributed to that group were produced by one individual, L26, a monolingual urban speaker. The other apparent cluster is among the older Harris women, though here all 11 tokens in that group were produced by one individual, H09. The other speaker in

the older female Harris group did not glottalize at all. Finally, monolinguals glottalized at significantly higher rates than bilinguals (Figure 7d; estimate = 3.628, $z = 9.975$, $p < 0.001$) at all age groups and in both male and female speakers.

Table 6: Distribution of glottalized tokens according to island, sex, age group, and place of articulation.

	Lewis		Harris		Skye		Total
	M	F	M	F	M	F	
Age 1 (17–25)							
p	2	6	0	1	3	7	19
t	1	22	0	5	18	40	86
k	1	7	0	1	9	17	35
Total	4	35	0	7	30	64	140
%	1.40	10.23	0.00	6.14	13.16	37.43	
Age 2 (35 -45)							
p	1	3	1	2	5	1	13
t	4	2	1	3	12	16	38
k	2	1	0	2	6	15	26
Total	7	6	2	7	23	32	77
%	25.15	2.11	0.88	6.14	1.32	0.44	
Age 3 (55+)							
p	0	0	0	2	12	9	23
t	2	2	2	9	29	23	67
k	0	0	0	0	17	12	29
Total	2	2	2	11	58	44	119
%	0.01	0.88	1.75	9.65	25.44	11.03	
						Total:	336

4.3 Analysis of devoicing

A commonly noted feature of HE is the absence among some speakers of voicing in obstruents, producing (near) mergers of /p t k/ ~ /b d g/, /tʃ/ ~ /dʒ/, and /s/ ~ /z/ (Wells 1982, Shuken 1984, Jones 2002, Stuart-Smith 2008). More rarely, a reversal of voicing is remarked; e.g., /s/ is realized as [z]. The usual assumption is that this trait has been acquired from the local varieties of Gaelic, which also lack a voicing contrast in such consonants, though contrast is maintained among stops via aspiration or preaspiration.

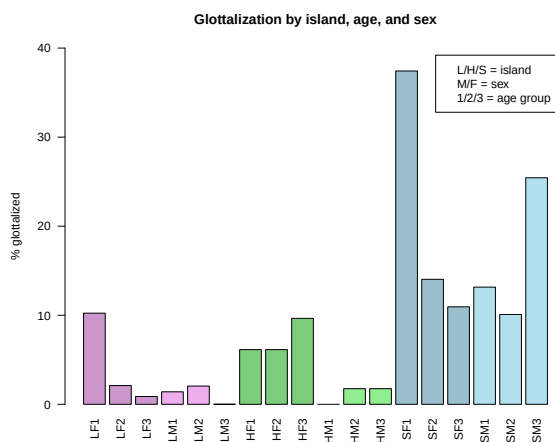


Figure 6: Comparison of glottalization rates by island (Lewis, Harris, Skye), sex, and age group (1 = 17–25, 2 = 35–45, 3 = 55+)

Table 7: Mixed-effects model of glottalization data. Intercept represents younger female bilingual speakers.

Random effects:		Variance	SD	
word		0.474	0.689	
Fixed effects:	Estimate	SE	<i>z</i>	<i>p</i>
(Intercept)	−3.971	0.349	−11.370	<0.001
sexM	−0.258	0.504	−0.513	0.608
age2	1.270	0.378	3.357	<0.001***
age3	0.922	0.377	2.445	0.014*
langM	3.628	0.364	9.975	<0.001***
sexM:age2	−1.270	0.640	−1.984	0.047*
sexM:age3	1.035	0.550	1.881	0.060
sexM:langM	−1.572	0.563	−2.794	0.005**
age2:langM	−3.791	0.511	−7.421	<0.001***
age3:langM	−2.088	0.468	−4.459	<0.001***
sexM:age2:langM	3.653	0.786	4.650	<0.001***
sexM:age3:langM	−16.378	33.615	−0.487	0.626

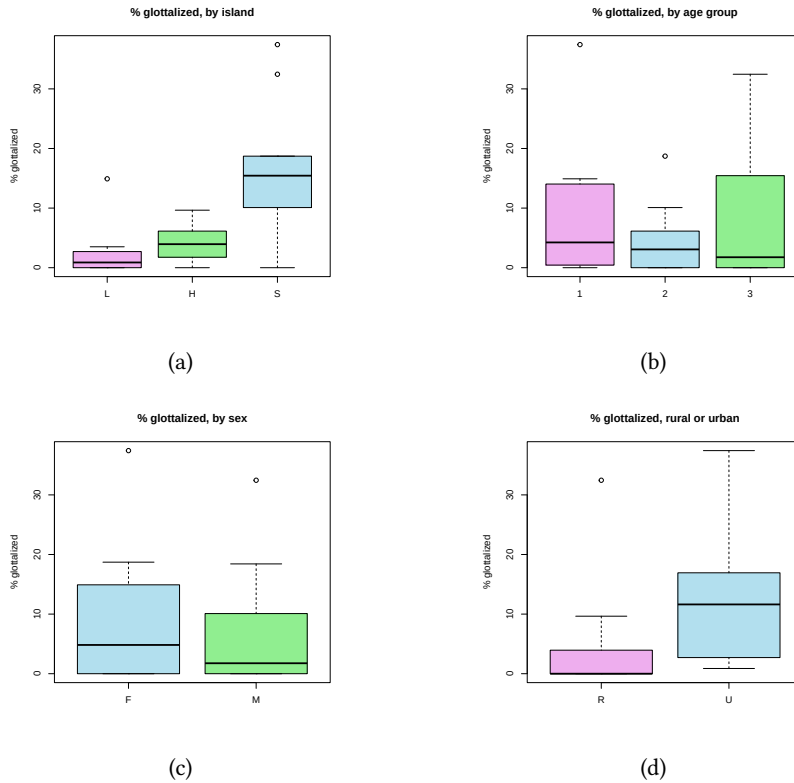


Figure 7: Glottalization rates by island (Lewis, Harris, Skye), age group (1 = 17–25, 2 = 35–45, 3 = 55+), sex, and rural vs. urban

While this lack of distinctive voicing has been noted to affect oral stops as well as sibilants in HE, Shuken’s collected material was particularly well designed to illustrate the feature among sibilants, especially in word-final position. Her stimulus set includes 33 potential contexts for devoicing, including 27 instances of word-final /z/, 1 example of /z/ in initial position; 1 medial and 2 final examples of /dz/; and 1 example of medial /ʒ/ (Table 3). There were thus 2310 potential occurrences of sibilant devoicing in the analyzed data. Of these, 639 tokens actually exhibited devoicing, identified as the absence of visible pulses or a voice bar in the waveform and spectrogram (Figure 8).

Devoicing was somewhat more equitably distributed across the demographic groups than either preaspiration or glottalization (Table 8). That said, there are clear patterns all the same (Figure 9, Figure 10). First, devoicing seems to be more frequent among Lewis speakers than among those from Harris or Skye

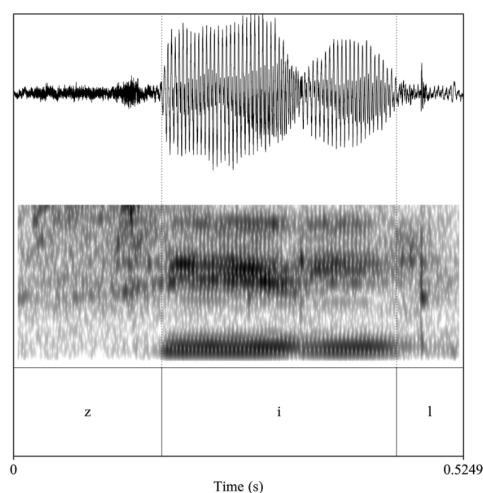


Figure 8: The word *zeal* as produced by H2, a rural bilingual female speaker from Harris in the 17–25 age group showing devoicing of the initial /z/.

(Figure 10a); more common in the young and middle age groups (Figure 10b); and more common among female than male speakers overall (Figure 10c).

Table 8: Distribution of devoiced tokens according to island, sex, age group.

	Lewis		Harris		Skye	
	M	F	M	F	M	F
Age 1 (17–25)						
Rural	39.4	56.1	36.4	22.7	na	na
Urban	29.3	44.7	na	na	15.2	38.4
Age 2 (35–45)						
Rural	45.5	48.5	15.2	21.2	na	18.2
Urban	24.2	53.5	na	na	4.5	39.4
Age 3 (55+)						
Rural	59.1	43.9	16.7	19.7	1.5	19.7
Urban	15.9	25.8	na	na	1.5	14.5

Statistical analysis of the devoicing data (Table 9) was conducted using a logistic mixed-effects model via the `glmer` function of the `lme4` package. The final

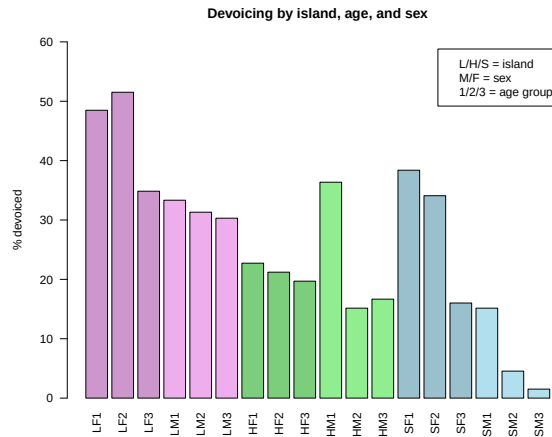


Figure 9: Rates of devoicing by according to island (Lewis, Harris, Skye), sex, and age group (1 = 17–25, 2 = 35–45, 3 = 55+)

model included *word* as random effect; and *island*, *age group*, *sex*, and *rural/urban* as fixed effects. Interactions between the fixed effects were also modeled. Overall, the model indicated that the effect of *island* was significant: the rate of devoicing was significantly higher in Lewis than in Harris (estimate = -1.755 , $z = 4.195$, $p < 0.001$), though not in Skye (estimate = -0.874 , $z = -1.360$, $p = 0.174$). We also see a general pattern of male speakers devoicing at significantly lower rates than female speakers in Lewis and in Harris, though not on Skye except among the older urban groups. The Skye result may be distorted by the absence of rural male Skye speakers in the younger and middle age groups, however.

5 Discussion

5.1 Summary of results

This analysis of a broad cross section of the recordings in the Shuken archive allows for a more detailed view of HE than was previously available. With respect to preaspiration and glottalization, this study both expands on Shuken (1984, 1985, 1986) and provides a baseline for comparison with studies using data collected several decades later (Clayton 2015, 2017, 2018). For example, in Clayton’s later study, preaspiration could be found at least sporadically in most HE speakers, including those from islands not included in Shuken’s sample. But of those speakers, only a small number of older rural women from Lewis provided more than a

Table 9: Mixed-effects model of complete devoicing results. Initial intercept represents younger female Lewis speakers.

Random effects		Variance	SD	
word		1.217	1.103	
Fixed effects	Estimate	SE	<i>z</i>	<i>p</i>
(Intercept)	0.247	0.337	0.733	0.464
islH	-1.755	0.418	-4.195	<0.001***
islS	-0.874	0.643	-1.360	0.174
age2	-0.376	0.388	-0.970	0.332
age3	-0.600	0.388	-1.544	0.123
rurbU	-0.562	0.337	-1.669	0.095
sexM	-0.826	0.390	-2.115	0.034*
islH:age2	0.276	0.592	0.466	0.641
islS:age2	-0.821	0.845	-0.972	0.331
islH:age3	0.394	0.596	0.661	0.509
islS:age3	-0.486	0.482	-1.010	0.312
islS:rurbU	0.560	0.570	0.981	0.327
age2:rurbU	0.812	0.488	1.664	0.096
age3:rurbU	-0.404	0.528	-0.765	0.444
islH:sexM	1.600	0.573	2.793	0.005**
islS:sexM	-2.485	1.650	-1.506	0.132
age2:sexM	0.677	0.548	1.235	0.217
age3:sexM	1.580	0.553	2.858	0.004**
rurbU:sexM	0.028	0.496	0.057	0.954
islS:age2:rurbU	0.436	0.852	0.512	0.609
islH:age2:sexM	-1.910	0.841	-2.271	0.023*
islS:age2:sexM	-0.724	0.732	-0.988	0.323
islH:age3:sexM	-2.583	0.842	-3.066	0.002**
islS:age3:sexM	-1.185	1.199	-0.988	0.323
islS:rurbU:sexM	1.845	1.585	1.164	0.244
age2:rurbU:sexM	-1.411	0.701	-2.012	0.044*
age3:rurbU:sexM	-1.467	0.743	-1.975	0.048*

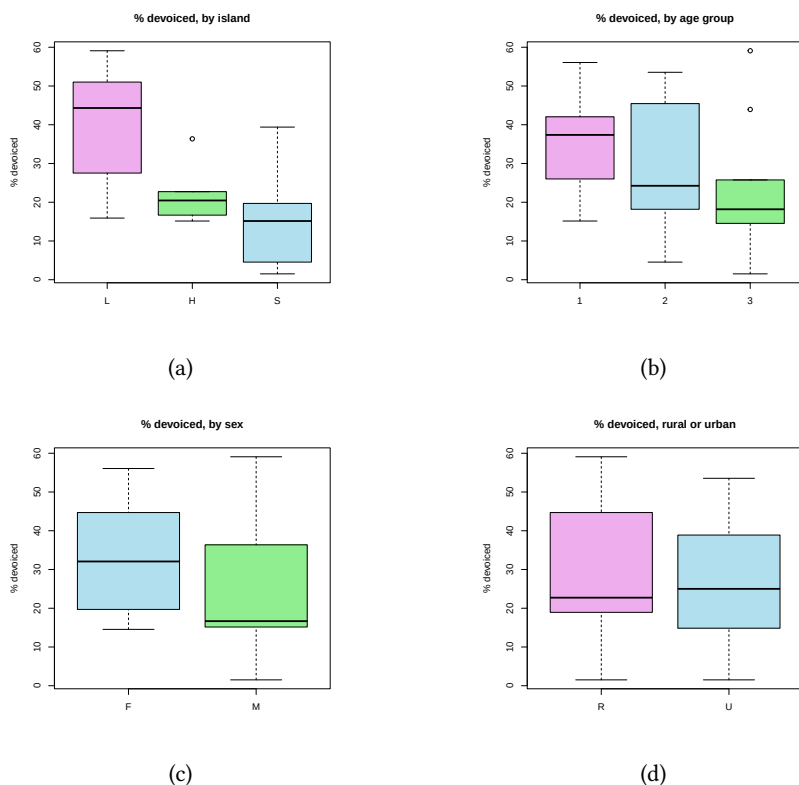


Figure 10: Devoicing rates by island, age group, sex, and rural vs. urban.

handful of preaspirated tokens. This could well be a development of the situation that Shuken (1985) discerned three decades previously, in which preaspiration predominated among participants from Lewis, particularly rural speakers, and particularly women. Though in that paper, Shuken did not consider the older two age groups from her full sample, the present analysis adds those two older age brackets, and finds that the same pattern holds: preaspiration is more abundant in Lewis than Harris or Skye, more abundant in rural areas compared to urban areas, and more abundant in women than men. The current study also finds that older speakers preaspirate more often than younger speakers. If we fast-forward three decades, so to speak, it is easy to see how the situation visible in Shuken's earlier data could readily develop into the state of affairs Clayton found in speakers three decades later.

Taken as a whole, the pattern we see with preaspiration is consistent with a conservative, local prestige feature, that is, one that speakers may exploit to

express affiliation with the local island community (cf. Labov 1963). Given the salience of preaspiration as a Gaelic feature, we may even suggest that the feature indexes a local Gaelic-speaking identity. However, with its evidently decreasing frequency, even among Lewis speakers, it is possible that preaspiration is becoming progressively less valuable to speakers in this regard.

On the other hand, the notably higher frequency of glottalization among Skye speakers, especially urban/monolingual speakers, instead bears the hallmarks of an incoming feature. Glottalization may well have been introduced, or reinforced, through contact with the mainland via newly arrived incomers from mainland urban communities like Glasgow or Aberdeen. Glottalization, particularly T-glottaling, is an expanding if stigmatized feature of working-class speech, in Scotland just as in England (Wells 1982, Trudgill 1988, Stuart-Smith 1999a,b). While we may expect a change favoring a non-standard form to be led by men (cf. Trudgill 1972, Labov 1963), especially one that confers working-class covert prestige, the lack of a significant difference by sex in the HE data is consistent with a similar lack of differentiation noted for T-glottaling in Glasgow English (Stuart-Smith 1999a), and indeed seems to presage a similar, later finding for HE in which glottalization shows no significant gender distinction (Clayton 2018).

Devoicing in certain respects patterns like preaspiration in the present study: it is more common among Lewis speakers than those from the other two islands (though not exclusive to Lewis), more common among female speakers than male speakers, and more abundant among rural speakers than urban speakers. On the other hand, devoicing is distinct from preaspiration in that it is noticeably more common among younger speakers than those in the middle or older age brackets (though statistically this age differentiation was only significant among men). The greater prevalence among women and among younger speakers is what we might associate with an incoming prestige feature (Labov 1990, Eckert 1989). But devoicing is not an incoming feature: it has been ascribed to the English of the Highland and Island region in one form or another for at least a century – sometimes represented as an anomaly that ought to be corrected (Grant 1913: 42), and sometimes caricatured as in Compton Mackenzie's (1947) novel *Whisky Galore*. Indeed, one might expect there to be a stigma associated with the feature, as there is with T-glottaling; in fact, during their interviews, at least two female speakers among Shuken's participants singled out devoicing as an incorrect or problematic aspect of the local English. Thus, devoicing is not a prestige feature in the conventional sense. One wonders, though, whether devoicing is being exploited by younger speakers in lieu of preaspiration: as a local prestige feature indexing islander identity.

5.2 Further directions

There are two broad areas of further analysis and fieldwork suggested by the analyses presented here. First, the findings discussed in this chapter prompt many questions about the current status of HE. The body of material collected for the most recent published study of HE (Clayton 2015, 2017, 2018) illustrated only preaspiration and glottalization, and so can tell us little about the third variable considered in this chapter, sibilant devoicing. There is also the issue of geographic scope. While Clayton did consider material from a relatively broad array of islands within the Hebrides island group, apart from Skye and Lewis most of those islands were represented by only one or two speakers; no speakers were from the more southerly islands in the archipelago. Mull, Jura, Islay, Barra, and many other islands thus remain completely uninvestigated.

There is also a substantial quantity of analysis remaining to be carried out within the Shuken collection, some of which is currently in progress. First, the large and varied stimulus set allows the consideration of additional phonological features beyond those already considered, for example retroflex /rs/ and vowel epenthesis (indeed, analyses of these two features are currently in progress). Second, the reading passage material should be added. As a stylistically distinct form of language use (cf. Trudgill 1972), this material could well bring additional focus to those variables considered in the present chapter and any future analyses. Finally, only a limited set of social factors were considered in this chapter, including island of origin, rural vs urban settings, sex, age, and language ability. While these factors have allowed a number of insights into the distribution of the linguistic features considered, in particular their geographic distribution and relative abundance within the relevant demographic groups, as yet we know little about how preaspiration, glottalization, and devoicing may correlate with social factors such as education, economic class, or profession; or how speakers may exploit them to express group affiliation or construct identity.

Acknowledgements

I would like to thank the very helpful staff at the University of Edinburgh School of Scottish Studies archives, in particular Elliot Holmes and Kirsty Stewart, who made this work possible. I would also like to thank the two anonymous reviewers for their very helpful comments and suggestions on a previous draft of this chapter.

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